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# Physical and Chemical Properties of Substances

All substances can be identified by the characteristics they possess.

Some of these characteristics can be observed without altering the chemical composition of the substance. These are called the **physical properties** of the substance. Physical properties may include:

> *colour, odour, boiling point, melting point, freezing point, electrical conductivity, thermal conductivity, magnetism, opacity, malleability, density and hardness.*

Some of these characteristics can only be observed when a substance is chemically transferred into another through a chemical reaction. These are called the **chemical properties** of the substance. Chemical properties may include:

> *flammability, heat of combustion, toxicity, reactivity with water, reactivity with oxygen, reactivity with acids, radioactivity, pH level and chemical stability.*

If a substance experiences a change in any of its physical properties, this is referred to as a **physical change**. Examples of physical changes include painting a wall to change its colour, melting gold to change its shape or running electricity through an electrical cord. During a physical change, the chemical composition of the substance does not change and no new substances are created. Physical changes are often reversible.

If a substance experiences a change through its chemical properties, this is referred to as a **chemical reaction**. Examples of chemical reactions include burning wood, frying an egg or the process of photosynthesis. During a chemical change, the chemical composition of the original substances changes and new substances are created. Chemical changes are often irreversible.



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# Physical and Chemical Properties of Substances – Sorting Task

**Task:**

| |
|---|
| In each of the following examples, a substance is undergoing change.<br>Decide whether the change occurring is a physical change or a chemical reaction.<br>Write the example into the correct column of the table below. |

*water boiling, a sliced apple turning brown, a nail rusting, stretching a piece of clay, heating a frying pan, burning a piece of wood, an ice cube melting, fireworks exploding, baking a cake, milk turning sour, digesting food, dissolving sugar in tea*

| Physical Changes | Chemical Reactions |
|---|---|
| | |



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# Physical and Chemical Properties of Substances – Answers

| Physical Changes | Chemical Reactions |
|---|---|
| *water boiling* | *a sliced apple turning brown* |
| *stretching a piece of clay* | *a nail rusting* |
| *heating a frying pan* | *burning a piece of wood* |
| *an ice cube melting* | *fireworks exploding* |
| *dissolving sugar in tea* | *baking a cake* |
| | *milk turning sour* |
| | *digesting food* |



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